

6G Weekly Survey — May 25 - May 31, 2026

3 items across 2 topics, ranked by importance.

Gen AI based Channel Modeling

CDiT: Conditional Diffusion Transformer for Geometry-Aware Terahertz Cross Far- and Near-Field Channel Generation

Zhengdong Hu, Chong Han · arXiv eess.SP · 2026-05-19

★★★★■ Addresses critical sub-THz/THz channel modeling challenges for 6G with a novel generative approach that handles near-field effects, backed by realistic validation and strong architectural choices combining diffusion models with transformers.

The paper proposes a conditional diffusion transformer (CDiT) for generating THz UM-MIMO channels that capture both far-field and near-field propagation characteristics. The approach formulates channel modeling as geometry-aware conditional generation in sparse beamspace, using position information to condition a diffusion-transformer architecture based on the hybrid planar-spherical wave model (HPSM). Experiments on realistic THz datasets show superior performance over benchmark methods in synthesizing high-fidelity spatially-dependent channels.

Contribution: First application of diffusion transformers to THz channel generation with explicit geometry conditioning for cross far-/near-field modeling, moving beyond sparsity assumptions to learn complex spatial channel distributions.

Relevance to your work: Directly relevant to your GAN-based generative channel modeling work and sub-THz waveform research, offering an alternative diffusion-based approach for THz UM-MIMO that explicitly handles geometry-dependent near-field effects.

6G Physical Layer

Hyperbolic Frequency Multicarrier Modulation for Wideband Linear Time-Varying Channels

Xuehan Wang, Jinhong Yuan, Jintao Wang, Zhi Sun, et al. · arXiv eess.SP · 2026-05-19

★★★★■ Addresses a genuine gap in high-mobility waveform design for wideband channels (relevant for NTN/LEO and vehicular 6G scenarios) with rigorous signal processing foundations, though experimental validation and comparison to state-of-art waveforms would strengthen impact.

This paper proposes Hyperbolic Frequency Multicarrier (HFMC) modulation for wideband linear time-varying channels where Doppler scaling (time dilation/contraction) from high mobility cannot be ignored. Inspired by AFDM, HFMC uses hyperbolic frequency modulated signals to absorb the Doppler scaling factor into an equivalent delay, enabling 1D characterization of doubly dispersive channels and exploiting multipath diversity. The authors establish approximate orthogonality conditions, analyze spectrum and capacity, and develop parameter optimization for efficiency-reliability tradeoffs.

Contribution: Novel waveform design that handles wideband Doppler scaling effects by transforming them into equivalent delays, extending AFDM principles to the passband wideband regime where constant Doppler approximations break down.

Relevance to your work: Directly relevant as a candidate 6G waveform addressing Doppler scaling in high-mobility scenarios (LEO direct-to-cell, high-speed vehicular), complementing your work on OTFS/AFDM and providing analytical framework for wideband LTV channel characterization.

DJSCC-Enabled Multi-User Semantic CSI Feedback for Hybrid Beamforming in Dual-Polarized cmWave Massive MIMO

Ziqi Han, Ziwei Wan, Hengwei Zhang, Keke Ying, et al. · arXiv cs.IT · 2026-05-19

★★★★ Solid incremental work combining existing DJSCC concepts with dual-polarization exploitation for practical cmWave systems, though the core semantic communication and task-oriented CSI feedback ideas are not fundamentally new.

The paper addresses CSI feedback overhead in dual-polarized cmWave massive MIMO-OFDM by proposing a DJSCC-based scheme where distributed encoders at UEs compress CSI semantically and the BS decoder directly optimizes hybrid beamforming without explicit CSI reconstruction. A cross-polarization interaction module exploits correlation between vertical and horizontal polarization channels to improve compression efficiency. Simulations show improved downlink sum rate under limited feedback budgets across various SNR conditions.

Contribution: Novel integration of DJSCC with semantic CSI feedback that bypasses explicit channel reconstruction, combined with a cross-polarization interaction module specifically designed for dual-polarized cmWave massive MIMO.

Relevance to your work: Directly relevant to your interests in CSI compression, massive MIMO, and semantic/task-oriented communication, demonstrating end-to-end learning for physical layer design in next-generation systems targeting 6G frequency bands.